Abdulmalik Abdulmawla

COMPUTATIONAL URBAN DESIGN · SPATIAL-ECONOMIC RESEARCH

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Spatial-economic researcher and computational urban designer. I decode raw urban data and parametrise the methods to explore it — GIS, network analysis, agent-based simulation, and data-driven tooling (Python, R, C#) — so that evidence-based decisions and designs come easily. My research asks where economic activity can locate in the city, and how urban space stays open for the small economies people build from below.

EDUCATION

2018 – present

PhD candidate — in progress · Computer Science in Architecture & Urbanism (InfAR), Bauhaus-Universität Weimar
Thesis: *"Spatial dependency of urban retail: detailed methods for positioning street-level activities"* — supervisor: Prof. Dr. Reinhard König

2013

Master of Arts in Architecture · Hochschule Anhalt, Dessau-Roßlau, Germany
Thesis: *"Heterogeneous-Atelier Industry"* (with Ali Farhan) — mapping Şişhane's informal atelier networks in Istanbul and simulating them as one bottom-up factory

2007

BSc Architecture & Engineering · Faculty of Engineering, Mansoura University, Egypt

EXPERIENCE

Dec 2023 – present

Research Associate — Chair of Transport System Planning
Bauhaus-Universität Weimar · Bauhaus Mobility / European Digital Innovation Hub Thuringia (EDIH.TH)
Collection and analysis of open mobility data; interactive data-exploration platforms; data-science consulting for public institutions, incl. training public administrations on European open-mobility-data standards and Mobilithek publishing; supervision of student theses (Digital Transport Research Lab).

2016 – 2023

Research Associate — Computer Science in Architecture (InfAR/InfAU)
Bauhaus-Universität Weimar
Teaching computational methods for urban planning and analysis; development of design and research tools (DeCodingSpaces Toolbox, mineR); master's-thesis supervision; conference research (Space Syntax Symposium, eCAADe) and support of academic publications.

2017 – 2019

Project Coordinator — Discovering City (DAAD exchange)
Bauhaus-Universität Weimar ↔ German-Jordanian University, Amman
Wrote the project proposal, managed the finances, organized the teaching and workshop programme.

2016 – 2020

Contributing Author — DeCodingSpaces Toolbox
DecodingSpaces GbR, Weimar
C# development for a Grasshopper toolbox for urban analysis; testing, documentation, consulting — incl. an interactive masterplan-analysis tool for the Blankenburger Süden development (client: TSPA, Berlin, 2019).

2014 – 2016

Research Associate — Teaching & Computational Design
Arab Academy for Science, Technology & Maritime Transport (AASTMT), Cairo
Computational and parametric design (Grasshopper, Processing, Arduino) in a RIBA Part I certified programme.

2012 – 2016

Co-founder & Architect — Gozour Studio
Mansoura, Egypt
Residential, education and competition projects (incl. Lusail Underpass Competition, Qatar).

2005 – 2011

Junior & Freelance Architect — Egypt & UAE
EAI Group, Abu Dhabi · Invert Studios, Cairo · Al Turath, Abu Dhabi
Early architectural practice: design, CAD, 3D modelling and visualization, client representation.

SELECTED PUBLICATIONS

2022 — Abdulmawla, A., Bielik, M., Schneider, S., & König, R. Mapping active frontages: a method for linking street-level activities to their surrounding urban forms — a case study of Weimar, Germany. *13th International Space Syntax Symposium (SSS13)*, Bergen.

2022 — Abdulmawla, A., Schneider, S., König, R., Bielik, M., & Fuchkina, E. Parametric urban data structuring and spatial query. *40th eCAADe*.

2019 — Bielik, M., König, R., Fuchkina, E., Schneider, S., & Abdulmawla, A. Evolving configurational properties: simulating multiplier effects between land use and movement patterns. *12th International Space Syntax Symposium (SSS12)*.

2018 — Abdulmawla, A., Schneider, S., Bielik, M., & König, R. Integrated data analysis for parametric design environments — mineR: a Grasshopper plugin based on R. *36th eCAADe*.

2017 — König, R., Bielik, M., Knecht, K., Abdulmawla, A., & Fuchkina, E. New methods for urban analysis and simulation with Grasshopper — the DeCodingSpaces-Toolbox. *CAAD Futures 2017*, Istanbul.

2017 — Schneider, S., Abdulmawla, A., & Donath, D. The effect of the street network on movement patterns and land use in small cities. *11th International Space Syntax Symposium*, Lisbon.

2017 — Dennemark, M., Schneider, S., König, R., Abdulmawla, A., & Donath, D. Towards a modular design strategy for urban masterplanning. *eCAADe 2017*, Rome.

2016 — Abdulmawla, A., & Farhan, A. Heterogeneous assemblages of atelier industries. *SimAUD 2016*, London [poster].

Full list: researchgate.net/profile/Abdulmalik-Abdulmawla

TEACHING & WORKSHOPS

University seminars: Algorithmic Architecture · Parametric Urban Design & Analysis · SYNCITY I & II · Circular Urbanism · Digital Traffic Simulation Lab · Open Mobility Data · Rent-a-Data-Scientist.
Workshops (organizer): Discovering Cities (GJU Amman, 2018) · Urban Analysis, Synthesis & Exploration with Grasshopper (eCAADe 2018, Łódź) · Rural-to-Urban Transformation (Emerging City Lab, Addis Ababa, 2016) · Simulating Agent-Based Systems using Processing (Cairo, 2015).

SKILLS

Programming: Python, C#, R, Grasshopper, Java/Processing, vvvv, Arduino

Spatial & analytical: GIS, network analysis, agent-based simulation, space syntax, spatial data engineering

Modelling & media: Rhino, AutoCAD, Revit, 3ds Max, Photoshop, Illustrator, InDesign

Languages: Arabic (native) · English (fluent) · German (very good)

RECOGNITION

2026 — Ten-year service recognition at the Bauhaus-Universität Weimar, presented by the University President.